

Who We Are



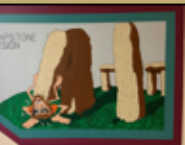
QL PLUS
Engineering Quality of Life for Injured Veterans

River
Deep
Foundation

- **Improved Fishing Device Controller**
- Ohm++
- Sponsored by Quality of Life Plus, partnered with River Deep Foundation

Garrett Hurd

- Electrical Engineer major
- Born and raised in the North Bay of California
- Admin Roles:
 - ✓ Team Lead
 - ✓ Schedule Manager
 - ✓ Backup for Assignments Manager
- Engineering Roles:
 - ✓ Power Management
 - ✓ Backup for System Architect
 - ✓ Backup for Analog Circuit Design



Tyler Martin

- Electrical Engineering Major
- Born in Hawaii, Raised in Colorado
- Admin Roles:
 - ✓ Customer Manager
 - ✓ Backup Communications Manager
 - ✓ Backup Moral Manager
- Engineering Roles:
 - ✓ Analog Circuit Designer
 - ✓ Test Engineer
 - ✓ RF Lead
 - ✓ Backup Mechanical Engineering Lead
 - ✓ Backup Power Management Architect



Pedro Mejido

- Electrical Engineer major
- Born and Raised in Texas
- Admin Roles:
 - ✓ SCRUM Master
 - ✓ Backup Budget Manager
- Engineer Roles
 - ✓ PCB Lead
 - ✓ System Architect



Felipe Tamayo Tapia

- Electrical and Computer Engineering major
- Born in Denver, Colorado
- Admin roles
 - ✓ Morale manager
 - ✓ Schedule manager
- Engineer roles
 - ✓ Embedded systems
 - ✓ Software engineer



Vedant Shah

- Electrical and Computer engineering major
- Admin Roles:
 - 1) Budget Manager
 - 2) Logistic Manager
 - 3) Backup Team lead
- Engineering Roles:
 - 1) Embedded Systems
 - 2) Software engineer
 - 3) User Interface

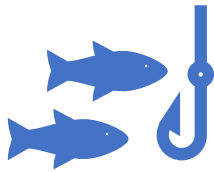


Travis Wood

- Electrical Engineer major
- From Washington State
- Admin Roles:
 - ✓ Communications Manager
 - ✓ Assignments Manager
 - ✓ Backup for Scrum Master and Customer Manager
- Engineering Roles:
 - ✓ User Interface
 - ✓ Mechanical
 - ✓ Backup for PCB and Testing



Product objectives and purpose of the system



Intuitive, fun, and easy to use fishing rod for people with spinal cord injuries and traumatic brain injuries (TBIs).



Used by patients of Craig Hospital during recreational therapy outings



Important behaviors and capabilities:

Easily used by spinal cord and TBI patients

Intuitive design, no manual required

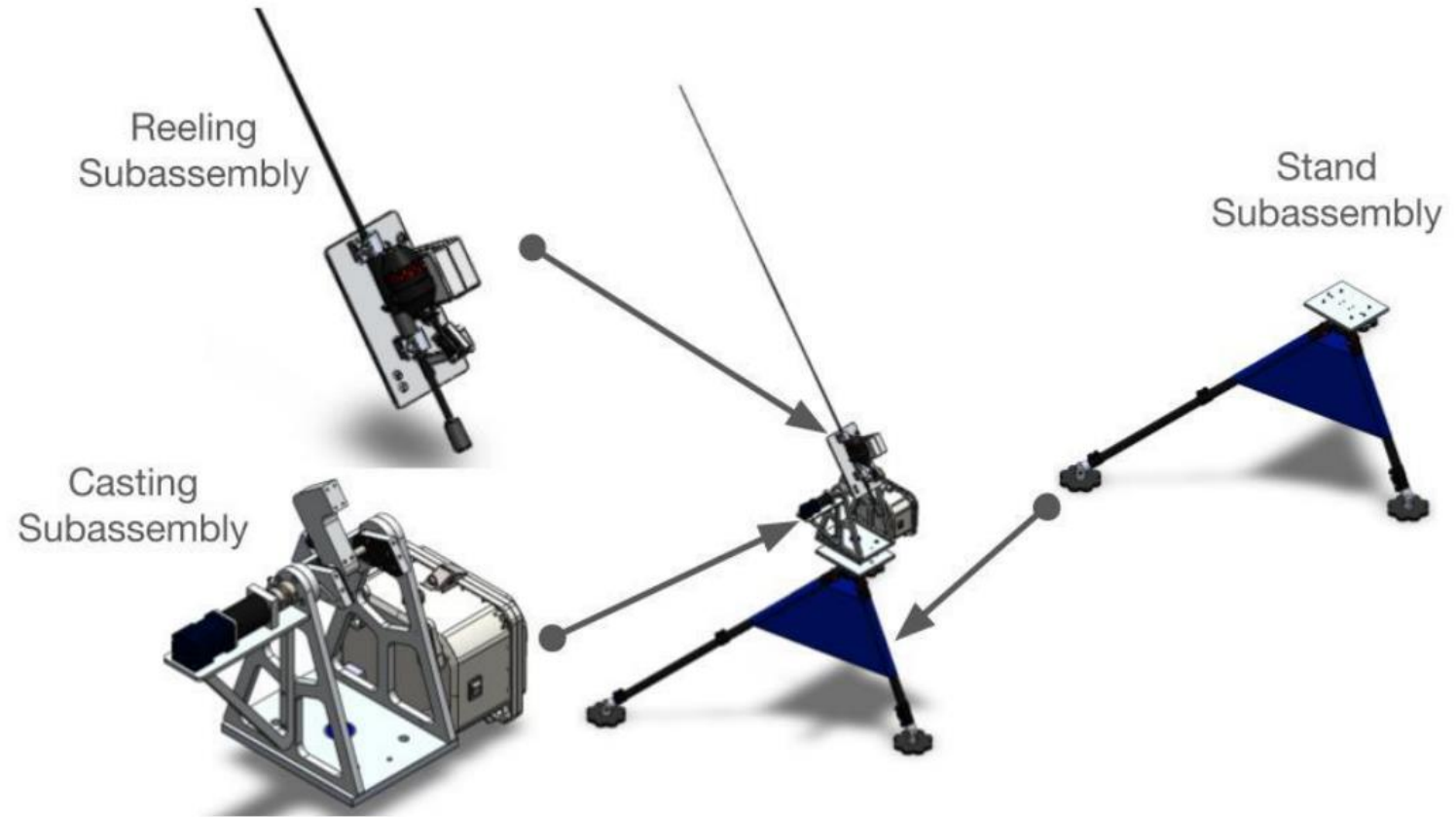
Two hours of battery life

Our number one priority is building the best user experience!

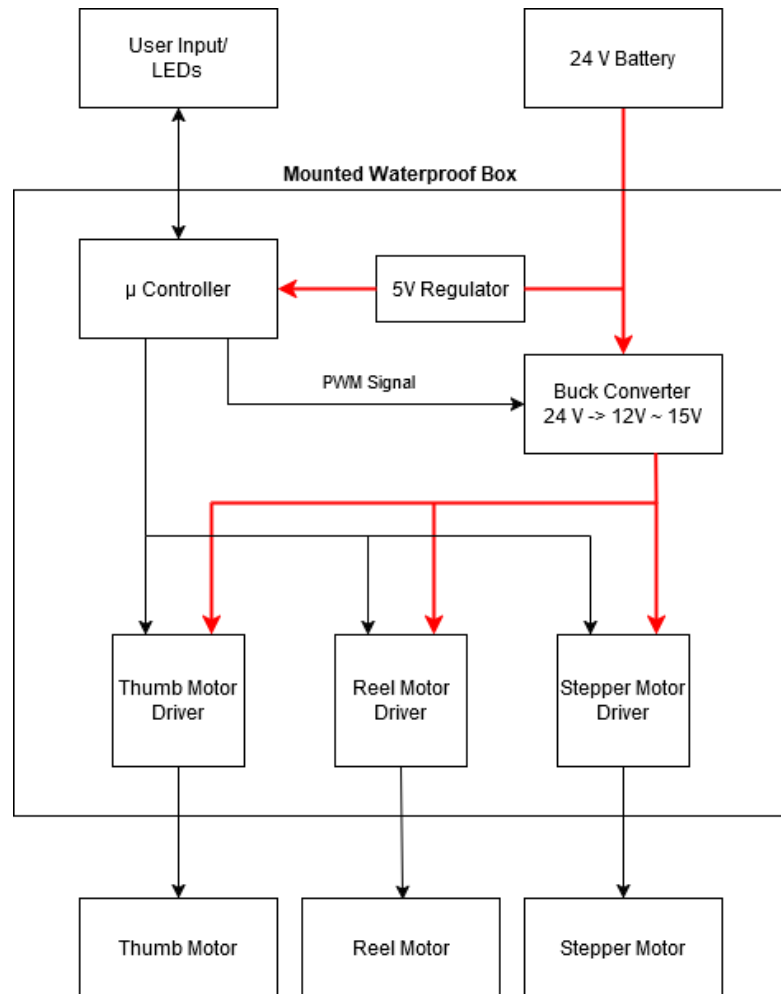
Overall Product Behavior

Overall Goal of Device:
Successfully cast out a
line, catch a fish and
reel in.

For Expo:
Demonstrate casting
and reel ability by
having user catch a
magnetic fish



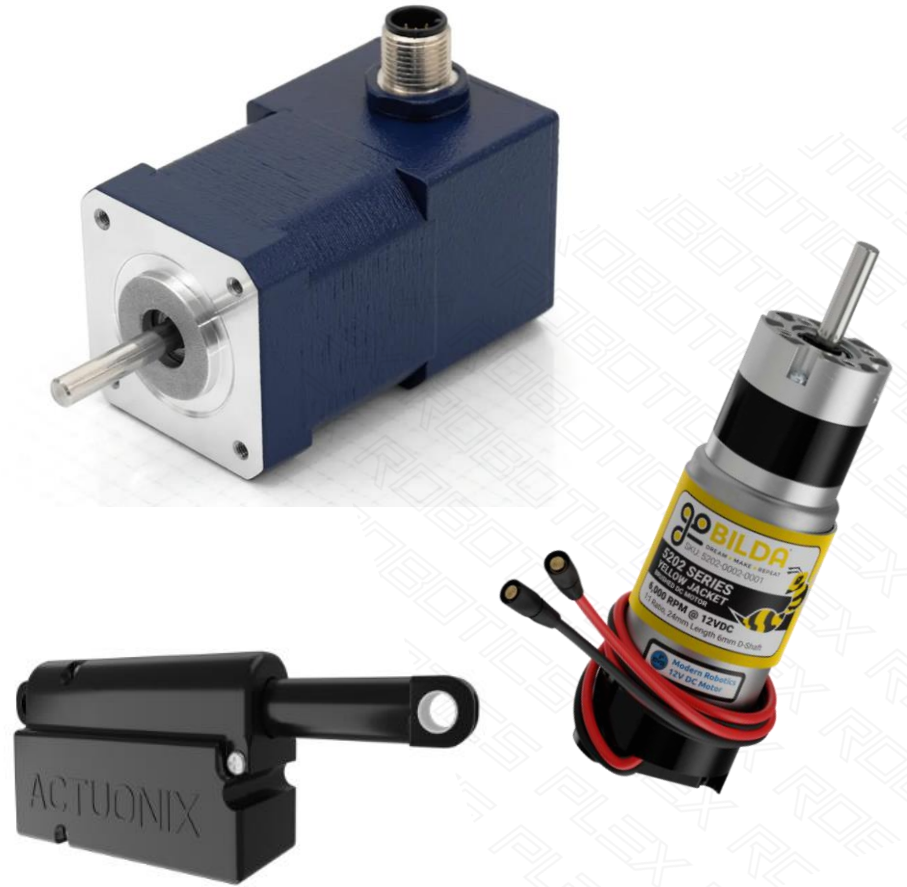
Overall Product Design



- Completed so far:
 - Removed old electrical system
 - Reworked controller design
 - Designed new electrical system
 - Created motor drivers
 - Updated operational flow of the system
- To be completed by Expo:
 - Incorporate custom PCBs/motor drivers
 - Incorporate new controller to system
 - UI testing for controller
 - Confirm power budget will last 2 hours
 - Stretch goal of bite tab controller

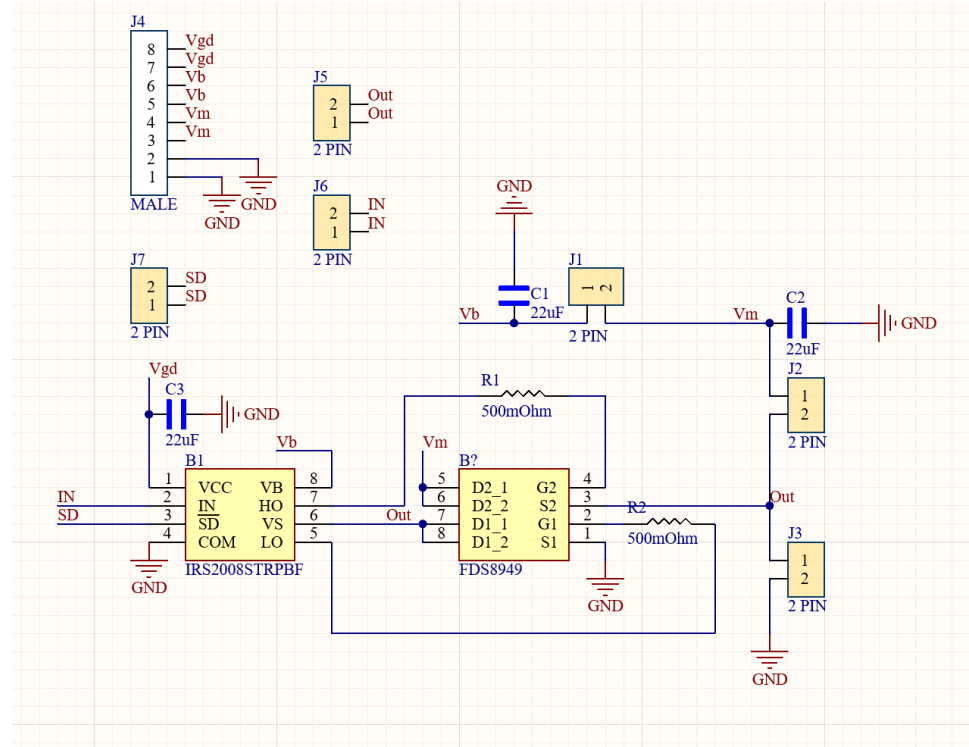
Motor Drivers

- Stepper Motor
 - ✓ Using a ATtiny85 to control a custom dual H-bridge
 - ✓ Uses NMOS transistors and gate drivers to drive a stepper motor at 12V and 7A
- Reel Motor
 - ✓ Will be using a singular NMOS transistor and be controlled with a PWM input
- Thumb Button Motor
 - ✓ Will use NMOS transistors and gate drivers to drive a solenoid

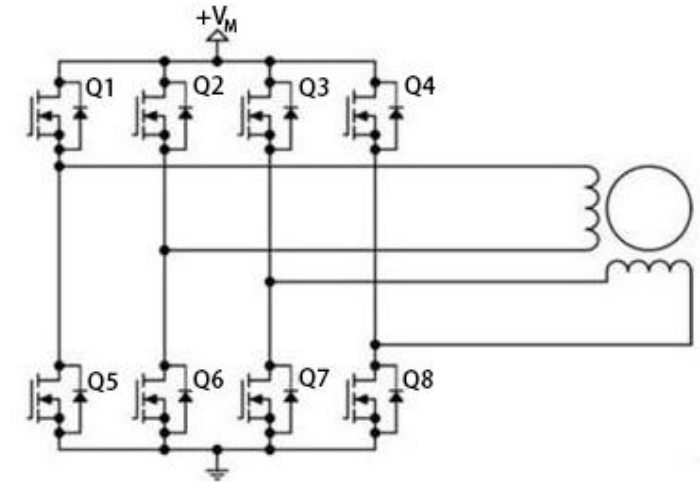


Stepper Motor

- Most complicated motor driver
- Using a IRS2008 Gate Motor Driver
- Using a FDS8949 Dual Power NMOS Transistor
- ATtiny85 Microcontroller



Current Half H-Bridge Design



Generic Bipolar Stepper Motor Driver Circuit From ATO.com

Technology options to achieve your product requirements

Microcontroller option: ATMEGA 328

Why:

- We are very familiar with it
- It's relatively inexpensive
- It's easy to use

Highest risk components:

Stepper Motor and Stepper Motor Driver

Why:

- Complex system

Technology option	Maturity risk	Complexity risk
ATMEGA 328	Low	Low
LEDs	Low	Low
Lever	Low	Low
Buck Converter	Medium	Low
Buttons	Low	Low
Stepper Motor	High	Medium
Stepper Motor Driver	High	Medium
DC Motor Driver	Low	Low
I/O Expander	Medium	Low
PID Speed Compensator	Low	Medium



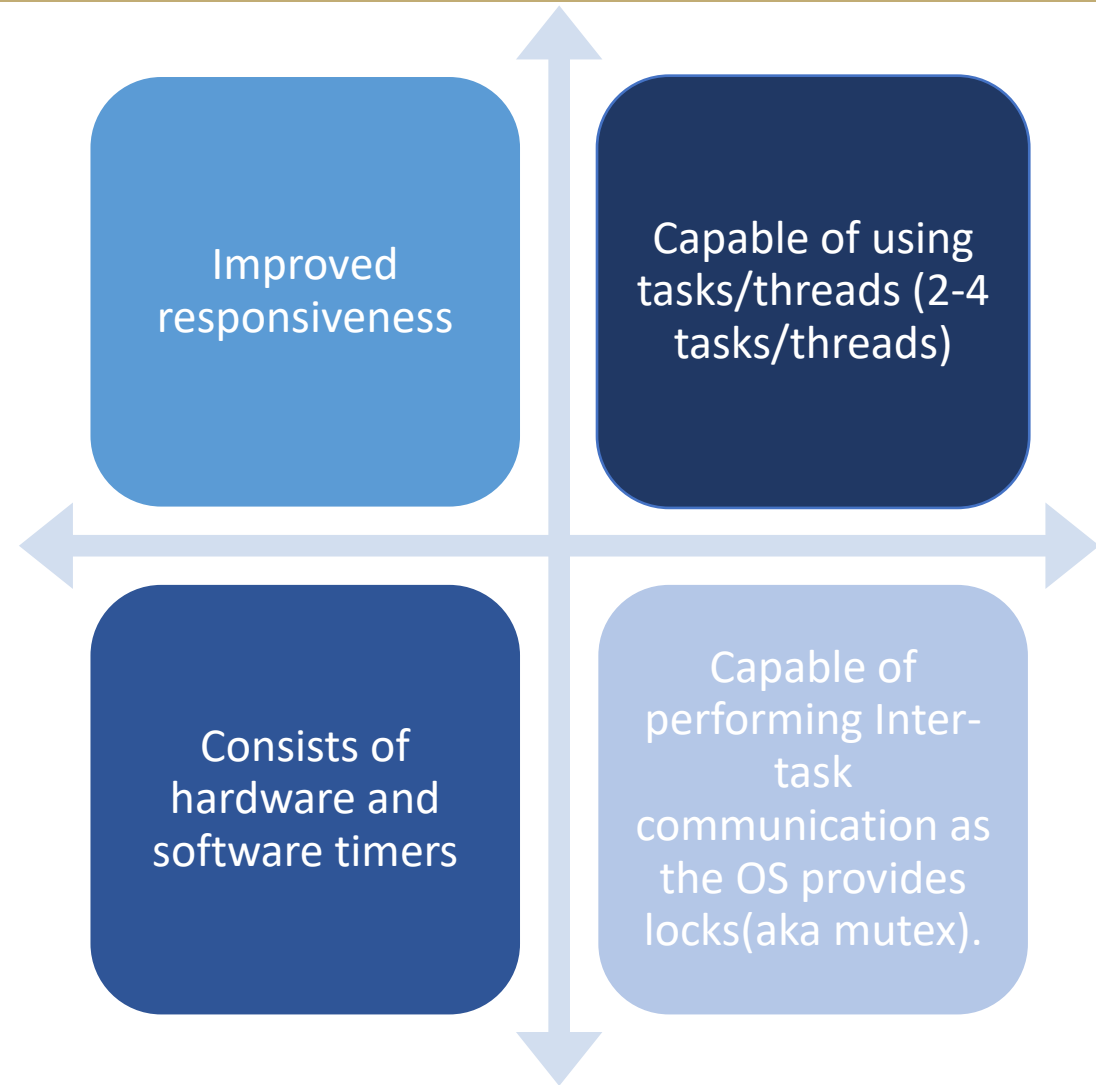
Using FreeRTOS and Arduino IDE



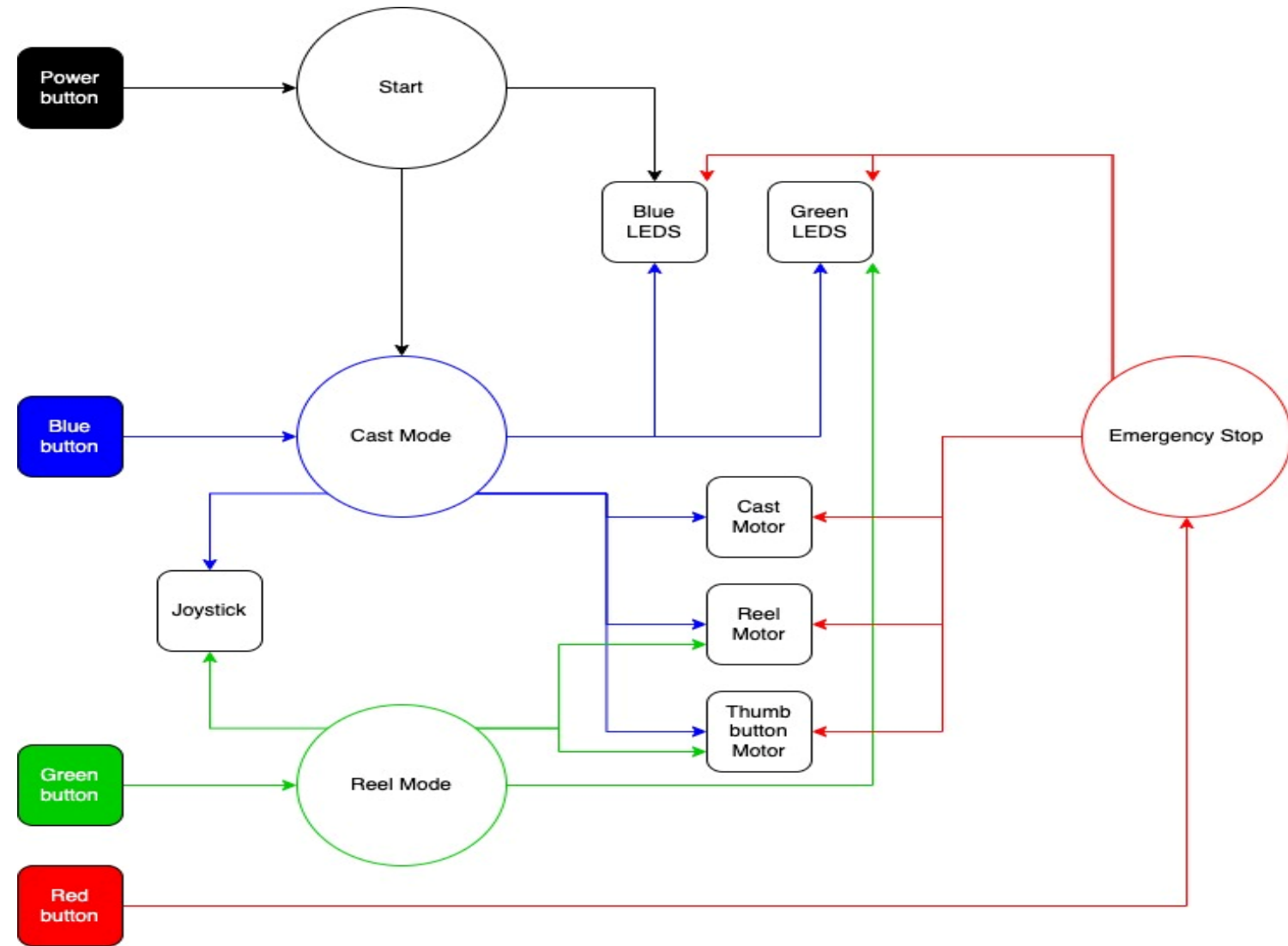
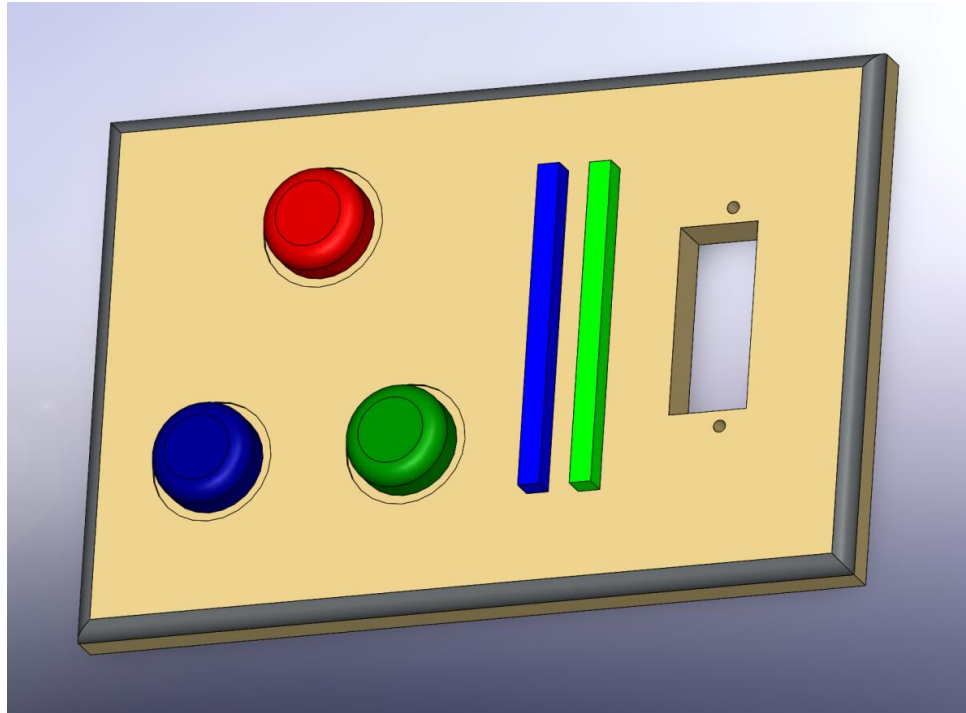
IDE



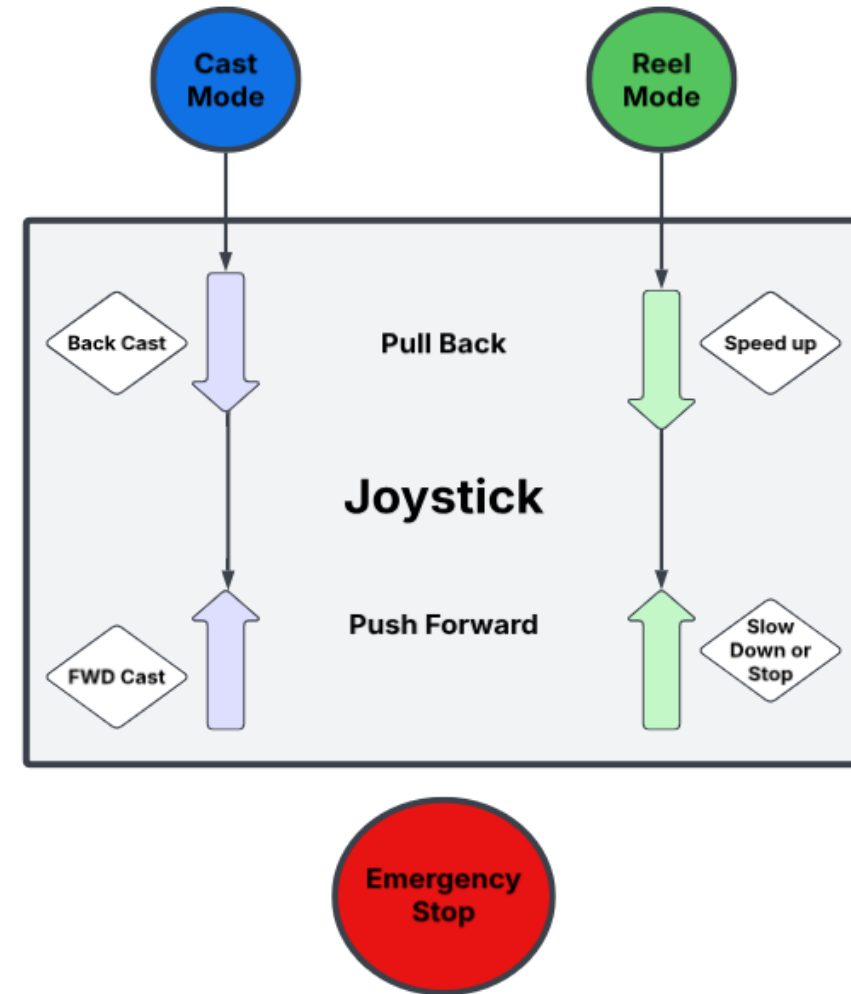
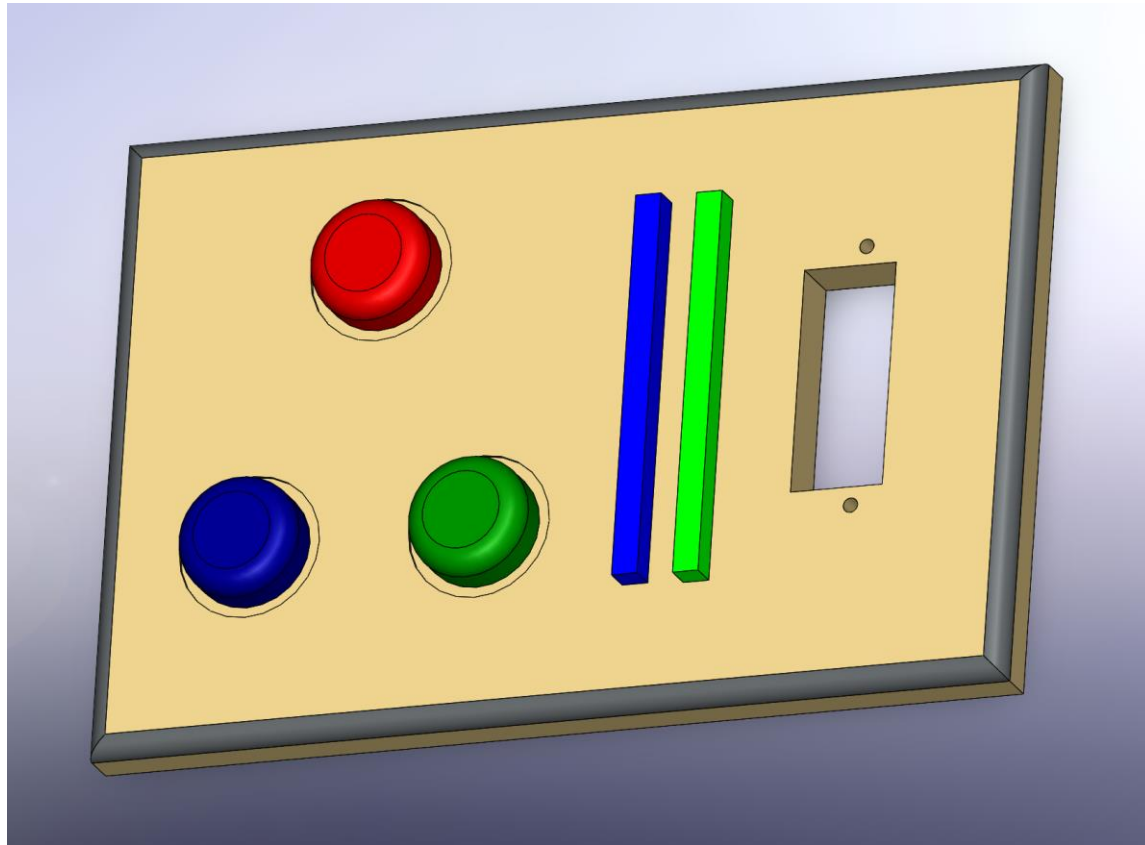
Operating System



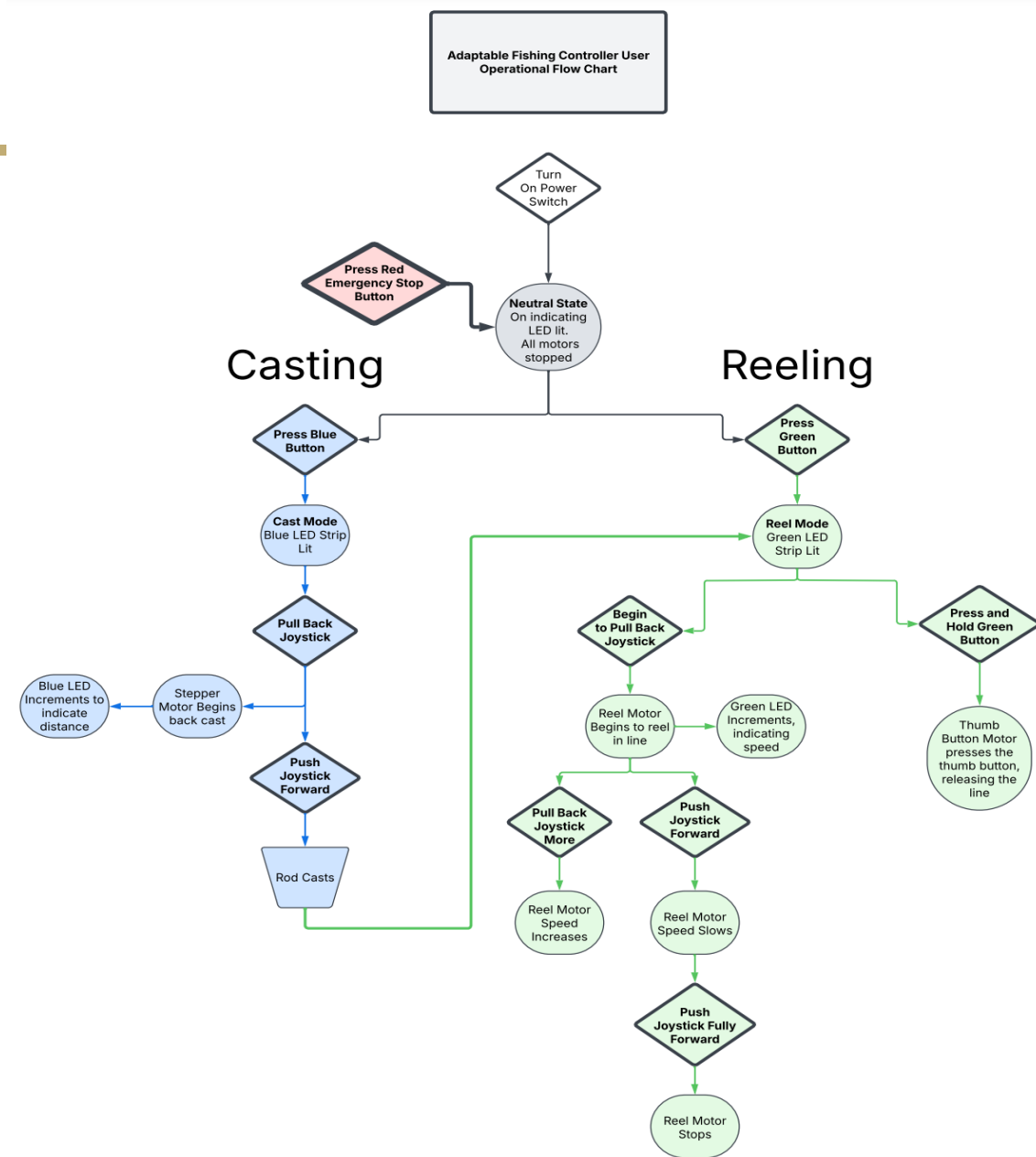
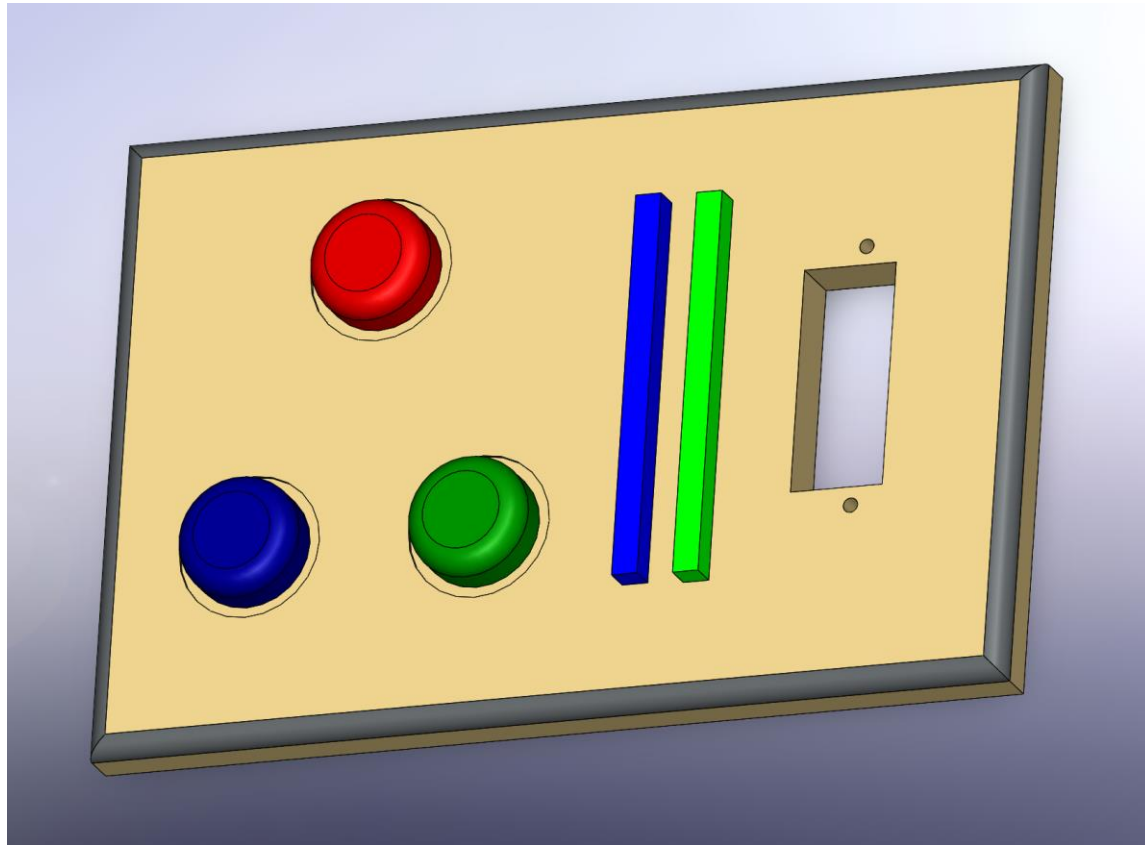
Basic Flow chart of software using FreeRTOS



Operational Flowchart



Operational Flowchart



Power Budget

- Input: 24V Lithium Battery by FLEX
 - ✓ 12 Ah, 2160 W
- Current calculations from Phase 1:
 - ✓ Casting Motor: Lower limit of 2A, max limit of 7A @ 12V, 0.16Ah
 - ✓ Reeling Motor: Lower limit of 0.41A, upper limit of 2A @12V, 0.0456 Ah
 - ✓ Thumb Motor: Lower limit of 0.2A, upper limit of 2A @ 12V, 0.00072Ah
- Current consumption for two-hour trip:
 - ✓ $2 * (0.16\text{Ah} + 0.0456\text{ Ah} + 0.00072\text{Ah}) = 0.4126\text{ Ah}$
- These values can change once we incorporate our custom hardware. This will be due to change in cast per hour, back cast speed, and reel speed.



Potential Software Risks and Mitigation

Risks:

- Microcontroller Resource Limitations: The ATmega328 has limited memory and processing power, potentially causing stack overflows.
- Unclear User Feedback & Error Handling: The system may not provide clear indications when an error occurs, leaving users confused.

Mitigation:

- Optimize code efficiency, minimize unnecessary tasks, and monitor stack usage.
- Implement LED indicators or simple auditory cues to signal operational status and errors.

Project Risks, Test Plans and Mitigation Strategies

- CONTROLLER MUST BE INTUITIVE
 - ✓ Sponsors number one concern and hardest to test
 - ✓ Feedback from large sample size on a variety of designs and inputs
 - ✓ Back up plan: make multiple controllers so that sponsor can use patients favorite
- Power Management
 - ✓ Ensure more than adequate power supply for longevity
 - ✓ Measure the output current from the battery while operating the fishing rod
 - ✓ Find ways to mitigate unnecessary power loss
- How much will we keep from the previous design and how much will we replace
 - ✓ Have to consider the maturity risk, complexity risk, time risk and financial risk of replacing components
 - ✓ Have decided to maintain motors, but completely redo all circuitry

Test Plan – Begin 4Mar25

- Cast distance – Integration Testing
 - ✓ Use a variety of weights on the line and back cast distance to measure cast distance to test reliability of casting. (10', 20', 30' +/- 10% for each cast)
 - ✓ Will look at how consistent the motor can cast by looking at the distribution of cast distance and the failure rate of casting over a very large sample size
- Test variable reel speed
 - ✓ Set reel in speed and test a variety of input speeds with a variety of weights on end of line
- User interface
 - ✓ Go to the business field and get feedback from random sample groups on how intuitive controller is, based on that feedback and time to up-and-running
 - ✓ Input from experts and end users – March once fishing opens up

Motor Test Plan

- Reel Motor
 - ✓ Test the relationship between motor speed and input duty cycle from Microcontroller
 - ✓ Will need to analyze how increased torque affects reel speed and power consumption
- Stepper Motor and Thumb Button Motor
 - ✓ Test the reliability of casting by running several casting trails to determine the accuracy of distance, the standard deviation of the distance, the failure rate of casting and any other problems that may arise
 - ✓ This task will be very repetitive and will require a lot of time to record enough data to get reliable information

Product Design Lessons Learned

- UI design requires multiple perspectives, inputs, and iterations
- Engineering decisions can change with new information (Controller housing design, motor control circuitry, battery design)
- Commenting code becomes more important as more people read or work on the same project
- Making adjustments later in the process causes cascading changes, illustrating the value of proper prior planning. (Battery design -> Housing changes)

Personal Lessons Learned

- Garrett: Lessons in leadership, managerial communications, delegation, and gaining confidence.
- Tyler: Teaching myself new information, learning to utilize resources and how to ask for help. Conquering the stepper motor, designing custom drivers.
- Felipe: Software and hardware restrictions, learning how to be more efficient in design decisions, working around library constraints. Created an RTOS that works efficiently.
- Vedant: Debugging, motor integration with software, better utilizing an OS. Learned how to research and develop more effectively.
- Pedro: Better understanding the time required for different tasks. Learning to open up to teammates and asking for help.
- Travis: Relearning 3D design, getting into the mindset of making engineering decisions. Good UI design lessons and polling for more perspectives.

Schedule

- Now to Week 7 (Feb 28th):
 - ✓ By the end of February, we will have implemented our custom hardware to our device to have it functioning again.
 - ✓ Once the device is functioning again, we can confirm our power budget and matches our requirements of the battery lasting two hours.
- Week 8 (March 4th) to Week 11 (March 29th):
 - ✓ Most of March will be doing UI test of the controller, ensuring that it matches with our sponsors requirements.
- Week 12 (April 1st) to Expo (April 25th):
 - ✓ April we will be refining and polishing the device for Expo. We will also be incorporating our stretch goal of a bite tab controller.

Budget Overview

- A total of \$1500 was allocated to us, out of which we have already spent around \$730.
- An additional \$250-300 is expected to be spent in the coming weeks
- At this point, we anticipate staying within budget without any concerns.
- Cost of non-commodity parts - \$140 for the electrical control enclosure.

Project Budget

Description	Quantity	Cost(\$)
Controller Casing*	1	140
Buttons	3	120
Hall effect fader (Lever)	1	120
Motor drivers	3	60
Smart LED*	1	30
Batteries	1	270
PCB	1	50
Bite Tabs (Stretch goal)*	1	80
Amphenol Connections*	1	150
Total		1020

Summary

Goals:

- Create an intuitive, rugged, and easy to use fishing controller for Craig Hospital patients with spinal cord injuries and TBI's, that is capable of use for a recreational therapy fishing outing.
- A functional remote-controlled fishing rod that can consistently cast a given distance
- Improve the power system so that it can be used for at least 2 hours, with a rate of 30 casts per hour.

Concerns:

- To test the user interface, we need to have a fully functioning remote-controlled fishing rod.

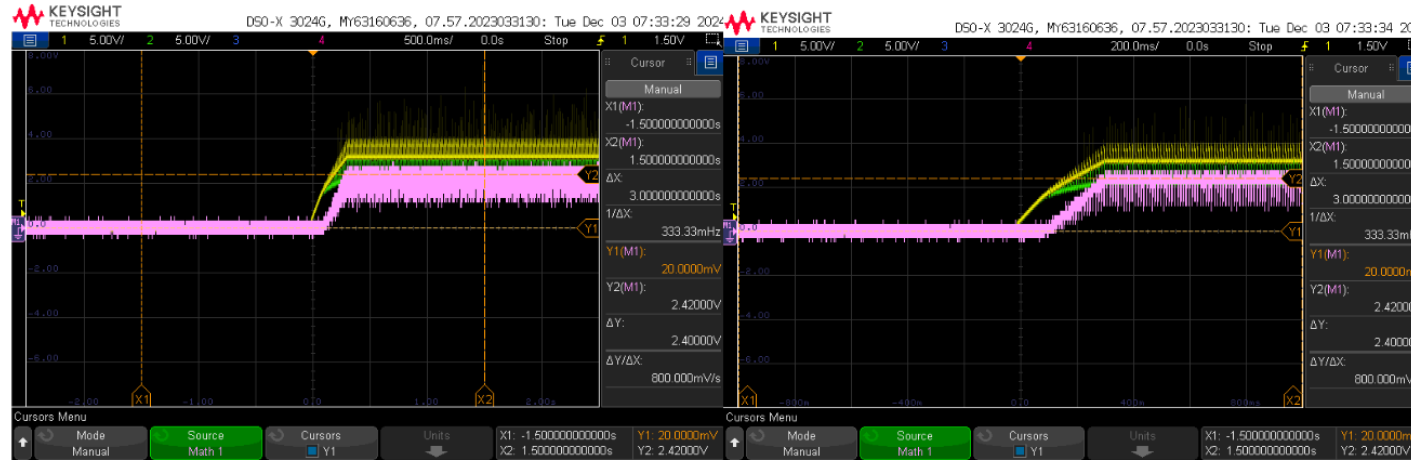


Thank You!

Questions?

Appendix I: Cast Motor Calculation

Casting Motor



$$I = \frac{2.4V}{1\Omega} = 2.4A$$

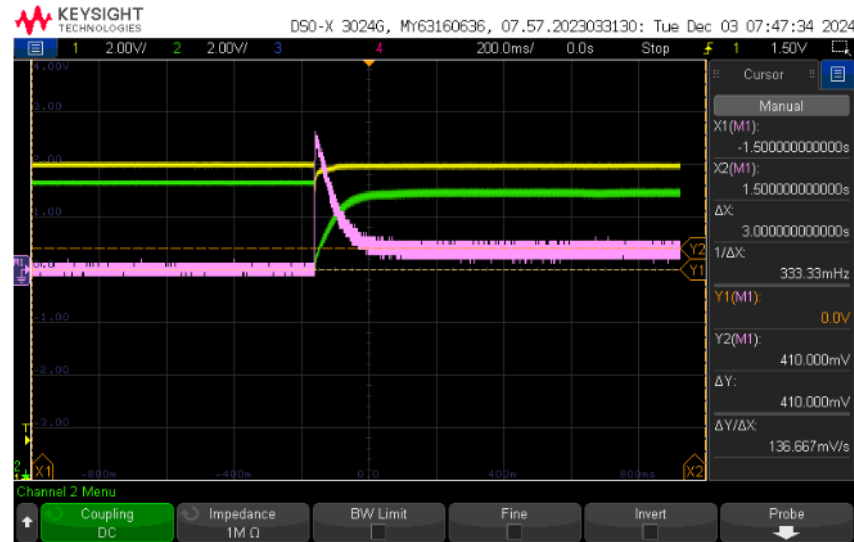
$$\text{Max Back Cast} = 12\text{seconds}$$

$$\left(\frac{20 \text{ Casts}}{\text{Hr}}\right) (12\text{seconds}) = 240\text{seconds} = 4\text{minutes} = 1/15 \text{ Hr}$$

$$4\text{minutes} @ 2.4A \Rightarrow \frac{2.4A}{15} = 0.16AH$$

Appendix II: Reeling Motor Calculation

Reeling Motor



$$I = \frac{410mV}{1\Omega} = 0.41A$$

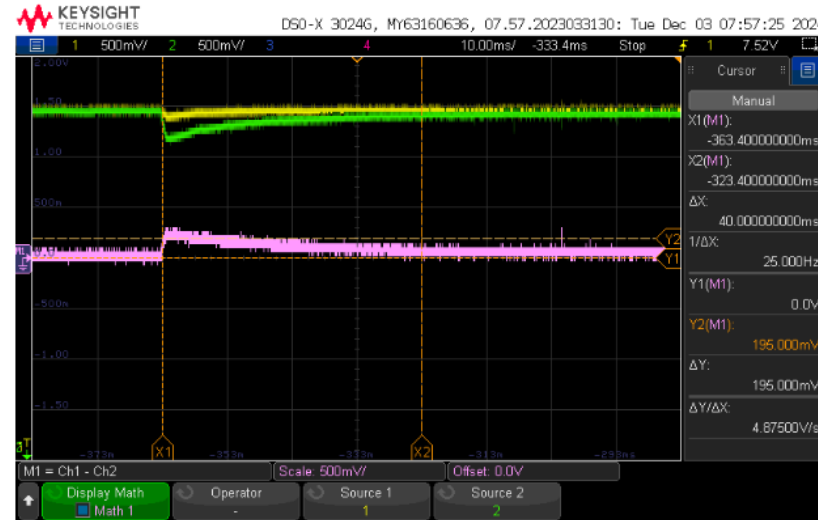
20ft Reel Time = 10 seconds $\Rightarrow$ 40ft ~20 seconds

$$\left(\frac{20 \text{ Casts}}{\text{Hr}}\right) (20 \text{ seconds}) = 400 \text{ seconds} = 6.66 \text{ Minutes} = 1/9 \text{ Hr}$$

$$6.66 \text{ minutes @ } 0.41A \Rightarrow \frac{0.41A}{9} = 0.0456AH$$

Appendix III: Thumb Motor Calculation

Thumb Button Motor



$$I = \frac{195mV}{1\Omega} = 0.195A$$

40 Actuations for 20 Casts @ 333.3mS = 13.3seconds/Hr

$$13.3 \text{ Seconds @ } 0.195A \Rightarrow \frac{0.195A}{270.67} = 0.00072AH$$